

Summary of the 5th International Workshop on Automated and verifiable Software sYstem DEvelopment (ASYDE) co-located with the 38th IEEE/ACM ASE 2023

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ABSTRACT

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Over the past three decades, automation in software development has gone mainstream. Software development teams strive to automate as much of the software development activities as possible, spanning requirements specification, system modeling, code generation, testing, deployment, verification, as well as release phases, project status reporting and system maintenance. Automation helps to reduce development time and cost, as well as to concentrate knowledge by bringing quality into every step of the development process. The Workshop on Automated and verifiable Software sYstem DEvelopment (ASYDE) provided a forum to share and discuss innovative contributions to research and practice related to novel software engineering approaches to automated and verifiable development of software systems. The 5th edition took place on September 11th, 2023, in Kirchberg, Luxembourg. Notably, this marked the inaugural co-location of ASYDE with the IEEE/ACM International Conference on Automated Software Engineering (ASE).

1. INTRODUCTION

Realizing high-quality software systems requires producing software that is efficient, error-free, cost-effective, and that satisfies evolving requirements. Thus, one of the most crucial factors impacting software quality concerns not only the automation of the development process but also the ability to verify the outcomes of each process activity and the goodness of the resulting software product as well. This becomes particularly true these days when we are, and will be, increasingly surrounded by a virtually infinite number of software artifacts – often underspecified – that can be composed to build new applications. This situation radically changes the way software will be produced and used: (i) software is increasingly produced according to a certain goal, that can change during the system's execution, and by integrating existing software; (ii) the focus of software production is then on the ability to perform automated reasoning to achieve software integration and development that can be kept always correct-by-construction via static and dynamic verification.

This calls for automated software development methods and techniques, compositional verification theories, integration architectures, as well as, automated flexible and dynamic composition, and development mechanisms.

Despite great interest in automated and verifiable software sys-

tem development, no common formal aspects and software engineering approaches have been fully established yet. Developing software systems via an automated generation and verification method encompasses a variety of foundational principles and practical aspects, ranging from modeling and analysis issues to model-checking, from model-driven development techniques and code synthesis to run-time management issues and machine learning tools. ASYDE'23 provided a forum to share and discuss innovative contributions to research and practice related to novel software engineering approaches to automated and verifiable development of software systems.

2. RELEVANCE TO ASE

The interplay between software engineering and foundational principles, as well as practical and formal aspects, is by nature tightly intertwined with automated and verifiable software system development. That is, in order to make software system development effective and elevate its maturity to the “readiness level” required for its adoption in practical software engineering contexts, novel formally grounded software engineering approaches, methods and tools are required, especially when automation of the system production and correctness of the system despite possible changes in the execution environment is of paramount importance. With this in mind, during the previous 4 years, ASYDE has been consolidating and strengthening its focus on automated software development.

For the above reasons, the IEEE/ACM ASE conference represented the most natural venue to host the 5th edition of ASYDE, a venue where researchers and practitioners had the possibility to meet, disseminate and exchange ideas, challenges and problems, further identifying and discussing key issues, sharing knowledge and experiences towards devising together possible solutions, how to automate them and how to synchronize on coming future efforts from automated software engineering perspectives.

3. SUMMARY OF THE 5TH EDITION

Paper format – ASYDE 2023 welcomed research papers, experience papers and tool presentations; nevertheless, papers describing novel research contributions and innovative applications were of particular interest.

Regular papers (up to 10 pages): In this category fell those contributions which proposed novel research contributions, addressed

challenging problems with innovative ideas, or offered practical contributions (e.g., industrial experiences and case-studies) in the application software engineering approaches for building software systems via automated development and verification. Regular papers clearly describe the situation or problem tackled, the relevant state of the art, the position or solution suggested and the potential benefits of the contribution. Authors of papers reporting industrial experiences were strongly encouraged to make their experimental results available for use by the reviewers and other researchers.

Short papers (up to 5 pages): This category included tool demonstrations, position papers, well-pondered and sufficiently documented visionary papers. Tool demonstration papers explain enhancements made in comparison to previously published work.

Workshop format – The 5th ASYDE consisted of a one-day workshop, on September 11th, with the final program of the workshop available online: <https://asyde-series.github.io/asyde2023/>. The primary aim of the ASYDE workshop was to encourage interesting discussions among attendees, by actively promoting new research ideas in the realm of automated and verifiable system development, and stimulating new collaborations among the participants. The starting point was a opening keynote followed by 3 paper sessions.

Keynote – The workshop program started with one keynote talk given by Maxime Cordy, research scientist at University of Luxembourg to stimulate discussions about automated machine learning security in the real world.

Paper Sessions – A sum of 18 submissions was received. Following an extensive peer-review process, which engaged three members from the program committee for each submission, a total of eight submissions were successfully selected for publication. Since the workshop was held in person all the presentations were delivered in person. A 30-minute time slot was allotted for each full-paper presentation (20 minutes for the actual presentation plus 10 minutes for Q&A), and a 20-minute session was set for each short paper (15 minutes for presentation and 5 minutes for discussion).

Presentations were organized according to the following sessions:

Session 1: AI and Intelligent Systems – This first paper session contained two short papers presentations: (i) “Exploring Early Adopters’ Perceptions of ChatGPT as a Code Generation Tool” [7], authored by Gian Luca Scoccia, and (ii) “Automated Negotiation - Preliminary results of a systematic mapping study” [5], authored by Mashal Afzal Memon, Gian Luca Scoccia and Marco Autili. Each paper presentation was followed by a general discussion involving the presenter, the session chair, and the other ASYDE participants.

Session 2: Prioritization and Optimization – The second paper session was about a mix of topics related to automatic test cases’ prioritization and system optimization. In particular, this session involved a short paper titled (iii) “ICO: A Platform for Optimizing Highly Configurable Systems” [4], authored by Édouard Guégain, Amir Taherkordi and Clément Quinton, and the following two long papers: (iv) “Test Case Prioritization based on Neural Network Classification with Artifacts Traceability” [6], authored by Ioana-Claudia Rotaru and Andreea Vescan, and (v) “Neural Network-based Test Case Prioritization in Continuous Integration” [8], authored by Andreea Vescan, Radu Găceanu and Arnold Szederjesi.

Session 3: Contract and Microservices – This final session was also about a mix of topics related to access control policies for smart contracts, automatic repairing of program specification, and automatic monoliths to microservices migration. Specifically, in this session, the following three papers were presented: (vi) “Modelling Multi-Party Role-Based Access Control Policies for iContractML Smart Contracts” [2], authored by Issam Al-Azzoni and Reiko Heckel, (vii) “Exploring Automatic Specification Repair in Dafny Programs” [1], authored by Alexandre Abreu, Nuno Macedo and Alexandra Mendes, and (viii) “Migrating from monoliths to microservices: enforcing correct coordination” [3], authored by Marco Autili, Gianluca Filippone and Massimo Tivoli.

Main topics and related takeaways – In summary, the main topics discussed during the workshop and related takeaways are:

- the use of ChatGPT for code generation can have a substantial impact in the automated software development landscape;
- the four main purposes of automated negotiation are: reward maximization, opponent modeling, preference elicitation, and agreement for cooperation;
- the ICO platform can be used in practice by developers to save time during the configuration and optimization of their software;
- behavior-driven development practices contribute to better test case prioritization techniques that require less human guidance;
- in continuous integration settings, the use of neural network-based classification models to help prioritize tests;
- authorisation requirements for multi-party role-based access control can be met by multiple parties across different individuals and roles;
- automatic corrections suggestion is a viable approach for developers to repair Dafny programs.

4. WORKSHOP HISTORY

The 5th Edition of the ASYDE workshop is the result of a follow-up action, being supported by the Steering Committee members, bringing together and consolidating the following previous events: OrChor at IEEE SERVICES’14, SCFI at IEEE SERVICES’15, SCART at SEFM’15, VeryComp at STAF’16, as well as the previous four editions of ASYDE at SEFM’19 to SEFM’22, three of which occurred during the COVID-19 pandemic.

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